DVLSI-2025 Assignment-5

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***Abstract*—This assignment involved schematic, symbol, layout, and characterization of standard-cell digital circuits including inverters, NAND, NOR, XOR2, MUX, mirror full adder, and D flip-flop.**

***Index Terms*—layout, characterization, FO4, Power, tpdr, tpdf**

I. Introduction

This assignment focuses on the design and analysis of basic standard-cell circuits such as inverters, NOR, NAND, XOR2, multiplexers, mirror full adder, and D flip-flop. The work includes schematic, symbol, layout creation, and characterization.

II. Layouts and Characterisation

FO4 propagation delay and switching power has been reported.

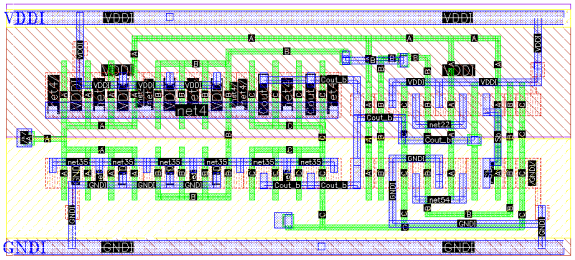


Fig. 1. Full Adder layout

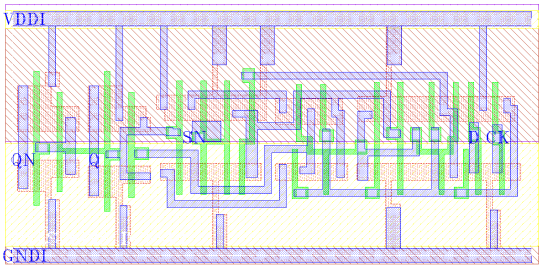


Fig. 2. DFFSX1 layout

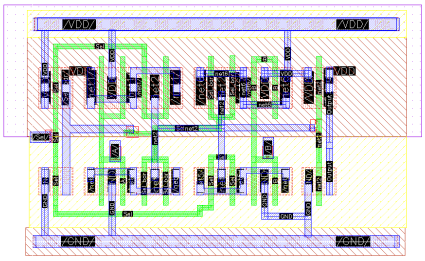


Fig. 3. MX2X1 layout

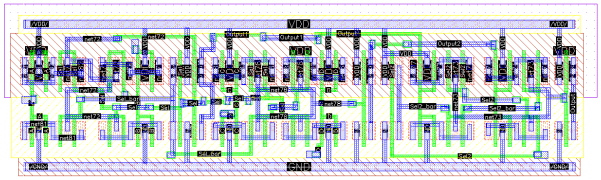


Fig. 4. MX4X1 layout

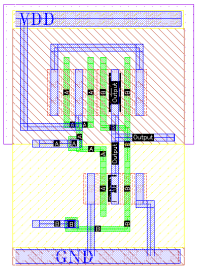


Fig. 5. NR2X1 layout

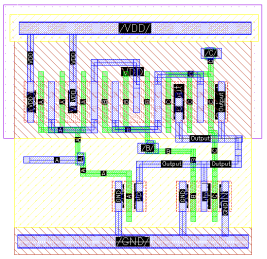


Fig. 6. NR3X1 layout

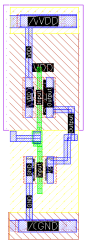


Fig. 7. INVX1 layout

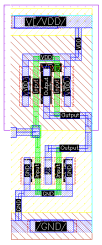


Fig. 8. INVX2 layout

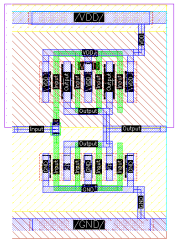


Fig. 9. INVX4 layout

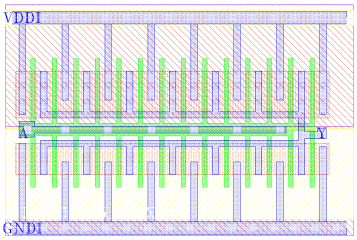


Fig. 10. INVX16 layout

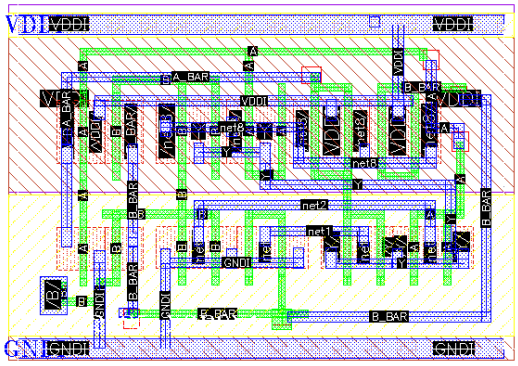


Fig. 11. XOR2X1 layout

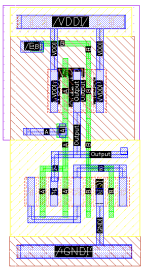


Fig. 12. ND2X1 layout

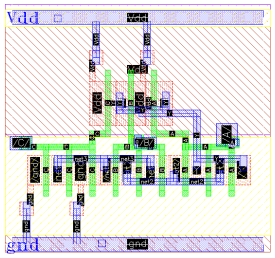


Fig. 13. ND3X1 layout

TABLE I
CHARACTERISATION

Cell	tpdr(ps)	tpdf(ps)	tpd avg(ps)	Power(μW)	Area(μm)^2
INVX1	26.5	27.5	27	8.585	1.05
INVX2	26.74	25.32	26.03	19.4631	1.6
INVX4	26.81	25.05	25.93	39.4408	3
INVX16	27.21	24.99	26.1	157.9227	6.4
NR2X1	26.85	18.04	22.445	17.2918	3.18
NR3X1	47.28	26.4	36.84	15.4257	4.89
ND2X1	23.42	15.9	19.66	14.6566	2.22
ND3X1	44.75	44.55	44.65	8.67897	4.45
MX2X1	24960	25050	25005	30.4613	5.83
MX4X1	84.71	86.81	85.76	30.7517	16.03
XOR2X1	907	1060	983.5	9.11	6
FA_D22	234.5	493.5	364	5.42789	9.86
DFFSX1	3839	178.2	2008.6	11.7889	9.04

III. CONCLUSION

In this assignment, a comprehensive mini standard cell library was successfully designed and characterized, including various logic gates, multiplexers, and flip-flops. The cells were verified for functionality, fitted to layout constraints, and documented with detailed characterization results. Overall, this library provides a solid foundation for system-level digital design, demonstrating key principles of standard cell methodology and layout optimization.

REFERENCES

[1] N. H. E. Weste and D. M. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, 4th ed., Addison Wesley, Pearson, 2011.